Model Methodology

Thermal Structure and Uncertainty Quantification
of Europa's Ice Shell

Full Mathematical Description

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Overview

This chapter presents the complete mathematical framework for modelling the thermal structure of Europa's ice shell. The model comprises five coupled components:

1. A **1D transient thermal solver** for the radial temperature profile through the ice shell (§2).
2. A **rheological model** coupling ice viscosity to temperature, grain size, and tidal forcing (§3).
3. A **parameterised stagnant-lid convection scheme** that partitions the shell into a conductive lid and convective sublayer (§4).
4. A **2D latitude-dependent extension** that couples independent radial columns through latitude-varying boundary conditions (§5).
5. A **Monte Carlo uncertainty quantification framework** with Bayesian observational constraining (§6).

Thermal Solver

Governing Equation

The model solves the one-dimensional transient heat conduction equation with volumetric tidal heating:

$$\rho(T)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{1}{\mathcal{G}} \frac{\partial}{\partial z} \left(\mathcal{G}\, k(T)\, \frac{\partial T}{\partial z} \right) + q_{\text{tidal}}(T) \tag{1}$$

where z is depth measured from the surface, $T(z,t)$ is temperature, and $\mathcal{G}$ is a geometric factor: $\mathcal{G} = 1$ in Cartesian coordinates (thin shells, $H < 30\,\text{km}$) or $\mathcal{G} = r^2$ in spherical coordinates (thick shells, $H \geq 30\,\text{km}$), where $r = R_p - z$ and $R_p = 1561\,\text{km}$ is Europa's radius.

Material Properties

Thermal conductivity

The thermal conductivity of ice I follows the Howell (2021) parameterisation:

$$k(T) = \frac{567}{T} \quad \text{W}\,\text{m}^{-1}\,\text{K}^{-1} \tag{2}$$

This inverse dependence on temperature is the single most important relationship in the model: it produces a factor-of-four conductivity contrast between the cold polar surface ($T \sim 50\,\text{K}$, $k \sim 11\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$) and the warm equatorial surface ($T \sim 110\,\text{K}$, $k \sim 5\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$).

Where near-surface porosity is present ($T < T_\phi = 150\,\text{K}$), the effective conductivity is reduced:

$$k_{\text{eff}}(T) = k(T)\,(1 - f_{\text{porosity}}) \quad \text{for } T < T_\phi \tag{3}$$

where $f_{\text{porosity}} \in [0, 0.10]$ is the porosity fraction.

Density

Temperature-dependent density follows Howell (2021):

$$\rho(T) = \rho_0\,[1 + \alpha_{th}(T_m - T)] \tag{4}$$

with reference density $\rho_0 = 917\,\text{kg}\,\text{m}^{-3}$ at the melting point $T_m = 273\,\text{K}$ and thermal expansion coefficient $\alpha_{th} = 1.6 \times 10^{-4}\,\text{K}^{-1}$.

Specific heat capacity

The specific heat capacity follows the temperature-dependent fit of Giauque & Stout (1936):

$$c_p(T) = 7.49\,T + 90.0 \quad \mathrm{J\,kg^{-1}\,K^{-1}} \tag{5}$$

yielding $c_p \approx 839\,\mathrm{J\,kg^{-1}\,K^{-1}}$ at $T = 100\,\mathrm{K}$ and $c_p \approx 2135\,\mathrm{J\,kg^{-1}\,K^{-1}}$ at the melting point.

Boundary Conditions

Surface boundary condition

The surface temperature is prescribed as a Dirichlet condition:

$$T(z = 0, t) = T_{\mathrm{surf}} \tag{6}$$

where T_{surf} is drawn from the Monte Carlo prior distribution (§6) or computed from the latitude-dependent profile (§5).

Alternatively, a radiative-equilibrium condition may be applied:

$$\varepsilon\,\sigma\,T_s^4 = (1-a)\,Q_\odot(\phi) + k(T_s)\,\left.\frac{\partial T}{\partial z}\right|_{z=0} \tag{7}$$

where $\varepsilon = 0.94$ is the surface emissivity, $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ the Stefan–Boltzmann constant, $a = 0.64$ the Bond albedo, and $Q_\odot$ the local solar flux. This nonlinear equation is solved via Newton–Raphson iteration using the linearisation $T^4 \approx 4T_0^3 T - 3T_0^4$.

Basal boundary condition: Stefan condition

At the ice–ocean interface ($z = H$), the temperature is fixed at the pressure-dependent melting point:

$$T(z = H) = T_m(H) = 273.0 - 7.4 \times 10^{-8}\,\mathrm{K\,Pa^{-1}} \times \rho\,g\,H \tag{8}$$

where $g = 1.315\,\mathrm{m\,s^{-2}}$ is Europa's surface gravity. For a 20 km shell, $T_m \approx 272.8\,\mathrm{K}$.

The shell thickness H evolves according to the Stefan condition governing the freezing front:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{q_{\mathrm{cond}} - q_{\mathrm{ocean}}}{\rho\,L} \tag{9}$$

where $L = 334\,000\,\mathrm{J\,kg^{-1}}$ is the latent heat of fusion, q_{ocean} is the heat flux delivered from the ocean to the base of the ice, and q_{cond} is the heat conducted upward from the base:

$$q_{\mathrm{cond}} = k(T_m)\,\left.\frac{\partial T}{\partial z}\right|_{z=H} \tag{10}$$

evaluated using a second-order one-sided finite difference:

$$\left.\frac{\partial T}{\partial z}\right|_{z=H} \approx \frac{3T_n - 4T_{n-1} + T_{n-2}}{2\,\Delta z} \tag{11}$$

If $q_{\mathrm{cond}} > q_{\mathrm{ocean}}$, the shell thickens (freezing); if $q_{\mathrm{cond}} < q_{\mathrm{ocean}}$, it thins (melting). The model is integrated to thermal equilibrium, defined as $|\mathrm{d}H/\mathrm{d}t| < 10^{-15}\,\mathrm{m\,s^{-1}}$.

Numerical Discretisation

Spatial discretisation

The radial domain $[0, H]$ is divided into n_z equally spaced nodes (default $n_z = 31$ for Monte Carlo production runs). The diffusion operator is discretised in flux-conservative form using harmonic-mean averaging of conductivity at half-nodes:

$$k_{i+1/2} = \frac{2\,k_i\,k_{i+1}}{k_i + k_{i+1}} \tag{12}$$

This ensures flux conservation across the conductive–convective interface where k_{eff} may jump by an order of magnitude when the Nusselt enhancement is applied (§4).

Time integration: Crank–Nicolson with Rannacher startup

The heat equation is advanced in time using the θ-method:

$$\rho\,c_p\,\frac{T_i^{n+1} - T_i^n}{\Delta t} = \theta\,\mathcal{L}[T^{n+1}] + (1-\theta)\,\mathcal{L}[T^n] + q_{\text{tidal}} \tag{13}$$

where $\mathcal{L}$ denotes the discrete spatial diffusion operator.

A Rannacher startup (Rannacher 1984) is employed: the first four half-steps use $\theta = 1$ (backward Euler) at $\Delta t/2$ to damp spurious oscillations arising from stiff initial conditions. All subsequent steps use $\theta = 0.5$ (Crank–Nicolson) at the full timestep $\Delta t = 10^{12}$ s ($\sim$31,700 years), giving second-order temporal accuracy.

This yields a tridiagonal system at each timestep:

$$a_i\,T_{i-1}^{n+1} + b_i\,T_i^{n+1} + c_i\,T_{i+1}^{n+1} = d_i \tag{14}$$

solved via the Thomas algorithm in $O(n_z)$ operations.

Picard iteration for nonlinearity

Because $k(T)$, $\rho(T)$, $c_p(T)$, and $q_{\text{tidal}}(T)$ all depend on temperature, the system is nonlinear. At each timestep, up to three Picard iterations are performed: material properties are evaluated at the current temperature estimate, the tridiagonal system is assembled and solved, and the new temperature is used to update properties. Convergence is declared when $\max_i |T_i^{(m+1)} - T_i^{(m)}| < 0.01\,\text{K}$.

Rheology and Tidal Dissipation

Composite Diffusion Creep Viscosity

The viscosity of polycrystalline ice is computed from the composite diffusion creep law of Howell (2021), which includes both volume diffusion and grain-boundary diffusion:

$$\eta(T, d) = \frac{1}{2}\left[\frac{42\,\Omega}{R\,T\,d^2}\left(D_v(T) + \frac{\pi\,\delta}{d}\,D_b(T)\right)\right]^{-1} \tag{15}$$

where $\Omega = 1.97 \times 10^{-5}\,\text{m}^3\,\text{mol}^{-1}$ is the molar volume of ice, $R = 8.314\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$ the gas constant, d the grain size, and $\delta = 7.13 \times 10^{-10}\,\text{m}$ the grain-boundary width. The diffusion coefficients follow Arrhenius laws:

$$D_v(T) = D_{0v}\,\exp\!\left(-\frac{Q_v}{R\,T}\right) \tag{16}$$

$$D_b(T) = D_{0b}\,\exp\!\left(-\frac{Q_b}{R\,T}\right) \tag{17}$$

with parameters listed in Table 1. Viscosity is clipped to the range $[10^{12}, 10^{25}]$ Pa s for numerical stability.

Table 1: Rheological parameters (Howell 2021).

Parameter	**Symbol**	**Value**	**Unit**
Volume diffusion prefactor	D_{0v}	9.1×10^{-4}	m^2/s
Boundary diffusion prefactor	D_{0b}	8.4×10^{-4}	m^2/s
Volume activation energy	Q_v	59,400	J/mol
Boundary activation energy	Q_b	49,000	J/mol
Molar volume	Ω	1.97×10^{-5}	m^3/mol
Grain-boundary width	δ	7.13×10^{-10}	m
Ice shear modulus	μ	3.3×10^9	Pa

Tidal Heating: Maxwell Rheology

Under the Maxwell (spring-dashpot) model, volumetric tidal dissipation is:

$$q_{\text{tidal}}^{\text{Max}}(T) = \frac{\varepsilon_0^2\,\omega^2\,\eta(T)}{2\left[1 + \dfrac{\omega^2\,\eta(T)^2}{\mu^2}\right]} \tag{18}$$

where ε_0 is the tidal strain amplitude, $\omega = 2.047 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}$ is Europa's orbital frequency, and $\mu = 3.3 \times 10^9$ Pa is the ice shear modulus. The factor of 1/2 arises from time-averaging $\sin^2(\omega t)$ over one orbital period.

Maxwell dissipation peaks sharply at $\omega\tau = 1$ where $\tau = \eta/\mu$ is the Maxwell relaxation time. Only a narrow temperature band satisfies this resonance condition; ice that is too cold ($\omega\tau \gg 1$, elastic response) or too warm ($\omega\tau \ll 1$, viscous flow) dissipates negligibly.

Tidal Heating: Andrade Rheology

The Andrade model (McCarthy et al. 2011; Renaud & Henning 2018) incorporates transient creep from sub-grain dislocation motion and grain-boundary sliding, producing broadband dissipation absent from the Maxwell model.

The complex compliance is:

$$J^*(\omega) = \underbrace{\frac{1}{\mu}}_{\text{elastic}} + \underbrace{\frac{\Gamma(1+\alpha)}{\mu}\,(\omega\tau\zeta)^{-\alpha}\,e^{-i\alpha\pi/2}}_{\text{Andrade transient creep}} - \underbrace{\frac{i}{\omega\,\eta}}_{\text{viscous}} \tag{19}$$

where $\alpha = 0.2$ is the Andrade creep exponent, $\zeta = 1.0$ is the normalisation factor, $\tau = \eta/\mu$ is the relaxation time, and Γ is the gamma function. The real and imaginary parts are:

$$J_{\text{re}} = \frac{1}{\mu} + \frac{\Gamma(1+\alpha)}{\mu}\,(\omega\tau\zeta)^{-\alpha}\,\cos\!\left(\frac{\alpha\pi}{2}\right) \tag{20}$$

$$J_{\text{im}} = \frac{1}{\omega\,\eta} + \frac{\Gamma(1+\alpha)}{\mu}\,(\omega\tau\zeta)^{-\alpha}\,\sin\!\left(\frac{\alpha\pi}{2}\right) \tag{21}$$

The dissipative (imaginary) part of the shear modulus is:

$$G'' = \frac{J_{\text{im}}}{J_{\text{re}}^2 + J_{\text{im}}^2} \tag{22}$$

Volumetric tidal heating under Andrade rheology is then:

$$q_{\text{tidal}}^{\text{And}}(T) = \tfrac{1}{2}\,\omega\,\varepsilon_0^2\,G''(\omega, T) \tag{23}$$

The Andrade term $(\omega\tau)^{-\alpha}$ decays as a gentle power law ($\alpha = 0.2$), producing significant dissipation across a much wider range of viscosities (and therefore temperatures) than the sharp Maxwell peak. This is the primary reason for adopting Andrade rheology: laboratory measurements on polycrystalline ice confirm broadband dissipation behaviour inconsistent with the single-timescale Maxwell model.

Stagnant-Lid Convection Parameterisation

Europa's ice shell is expected to convect in the stagnant-lid regime, where a rigid conductive lid overlies a slowly overturning convective interior. Rather than resolving convective flow, the model parameterises the heat transport enhancement using scaling laws from numerical experiments (Solomatov & Moresi 2000; Green et al. 2021; Deschamps & Vilella 2021).

Conductive–Convective Interface Detection

At each timestep, the model scans the temperature profile from the surface to find the critical rheological transition temperature T_c below which the viscosity contrast is too large for convective participation.

Interior temperature (Deschamps & Vilella 2021, Eq. 18)

$$T_i = B\left[\sqrt{1 + \frac{2}{B}(T_m - c_2\,\Delta T_s)} - 1\right] \tag{24}$$

where $B = Q_v/(2\,R\,c_1)$, $c_1 = 1.43$, $c_2 = -0.03$, and $\Delta T_s = T_m - T_{\text{surf}}$.

Viscous temperature scale (Green et al. 2021)

The viscous temperature scale ΔT_v characterises the temperature interval over which viscosity changes by one e-fold:

$$\Delta T_v = -\frac{\eta(T_i)}{\mathrm{d}\eta/\mathrm{d}T\big|_{T_i}} = \frac{R\,T_i^2}{A_v\,T_m} \tag{25}$$

where $A_v = Q_v/(N\,R\,T_m)$ and N is the stress exponent.

Lid-base temperature

The conductive lid base temperature is:

$$T_c = T_i - \theta_{\text{lid}}\,\Delta T_v \tag{26}$$

where $\theta_{\text{lid}} = 2.24$ (Green et al. 2021, following Deschamps & Vilella 2021).

Interface depth

The conductive lid thickness D_{cond} is obtained by scanning the numerical temperature profile for the first depth z_c where $T(z) \geq T_c$, with linear interpolation between grid nodes:

$$z_c = z_{j-1} + \frac{T_c - T_{j-1}}{T_j - T_{j-1}}\,(z_j - z_{j-1}) \tag{27}$$

The convective sublayer thickness is $D_{\text{conv}} = H - z_c$.

Rayleigh and Nusselt Numbers

The Rayleigh number of the convective sublayer is:

$$\mathrm{Ra} = \frac{\rho\, g\, \alpha_{th}\, \Delta T_{\mathrm{conv}}\, D_{\mathrm{conv}}^3}{\kappa\, \bar{\eta}} \tag{28}$$

where $\Delta T_{\mathrm{conv}} = T_m - T_c$, $\kappa = k/(\rho\, c_p)$ is the thermal diffusivity, and $\bar{\eta}$ is the viscosity evaluated at the mean convective temperature. Convection is active when $\mathrm{Ra} \geq \mathrm{Ra}_{\mathrm{crit}} = 1000$.

For supercritical flow, the Nusselt number is:

$$\mathrm{Nu} = C\, \mathrm{Ra}^{\xi} \left(\frac{T_i - T_c}{\Delta T_{\mathrm{conv}}} \right)^{\zeta} \tag{29}$$

with $C = 0.3446$, $\xi = 1/3$ (Solomatov & Moresi 2000), and $\zeta = 4/3$ accounting for internal tidal heating (Green et al. 2021). For subcritical conditions ($\mathrm{Ra} < \mathrm{Ra}_{\mathrm{crit}}$), $\mathrm{Nu} = 1$ (pure conduction).

Effective Conductivity

Within the convective sublayer, the thermal conductivity is enhanced by the Nusselt factor:

$$k_{\mathrm{eff}}(z) = \begin{cases} k(T(z)) & z < z_c \quad \text{(conductive lid)} \\ \mathrm{Nu} \cdot k(T(z)) & z \geq z_c \quad \text{(convective sublayer)} \end{cases} \tag{30}$$

For cold polar columns where convection may be marginal, a smooth ramp function prevents numerical discontinuities:

$$\mathrm{Nu}_{\mathrm{eff}} = 1 + \alpha_{\mathrm{ramp}}\, (\mathrm{Nu} - 1) \tag{31}$$

where α_{ramp} is a C^1-continuous smoothstep in T_{surf}:

$$\alpha_{\mathrm{ramp}} = \begin{cases} 0 & T_{\mathrm{surf}} \leq 60\,\mathrm{K} \\ 3t^2 - 2t^3 & 60\,\mathrm{K} < T_{\mathrm{surf}} < 80\,\mathrm{K} \\ 1 & T_{\mathrm{surf}} \geq 80\,\mathrm{K} \end{cases} \tag{32}$$

with $t = (T_{\mathrm{surf}} - 60)/(80 - 60)$.

Latitude-Dependent Extension

The 1D model is extended to a 2D axisymmetric configuration by coupling N_ϕ independent radial columns (default $N_\phi = 19$) at latitudes $\phi \in [0, \pi/2]$ (equator to pole), each with latitude-appropriate boundary conditions.

Surface Temperature Profile

The annual-mean surface temperature varies with latitude as:

$$T_s(\phi) = \left[(T_{\mathrm{eq}}^4 - T_{\mathrm{floor}}^4)\, \cos^p \phi + T_{\mathrm{floor}}^4 \right]^{1/4} \tag{33}$$

where $T_{\mathrm{eq}} = 110\,\mathrm{K}$ is the equatorial temperature, $T_{\mathrm{floor}} = 46\,\mathrm{K}$ is the polar floor (Ojakangas & Stevenson 1989), and $p = 1.25$ is an exponent calibrated against the seasonal energy balance model of Ashkenazy (2019), reducing the RMS temperature error from $3.01\,\mathrm{K}$ ($p = 1$) to $0.70\,\mathrm{K}$.

Tidal Strain Latitude Profile

Tidal strain varies with latitude following the eccentricity-tide dissipation pattern of Beuthe (2013). The default whole-shell (mantle–core) pattern gives monotonically increasing strain toward the poles:

$$\varepsilon_0(\phi) = \varepsilon_{\mathrm{eq}} \sqrt{1 + c \sin^2\phi} \tag{34}$$

where

$$c = \left(\frac{\varepsilon_{\mathrm{pole}}}{\varepsilon_{\mathrm{eq}}}\right)^2 - 1 \tag{35}$$

With $\varepsilon_{\mathrm{eq}} = 6.0 \times 10^{-6}$ and $\varepsilon_{\mathrm{pole}} = 1.2 \times 10^{-5}$ (a 2:1 pole-to-equator ratio; Tobie et al. 2003), $c = 3$.

Two alternative patterns are also implemented:

- **Shell-dominated** (thin-shell membrane): $\varepsilon_0(\phi) = \varepsilon_{\mathrm{pole}} \sqrt{1 + c' \cos^2\phi}$ — monotonically increasing toward the equator.
- **Non-monotonic** (degree-4 amplification): mantle–core $\times[1 + A_m \sin^2(2\phi)]$ — peak at mid-latitudes ($A_m = 0.3$ default).

Ocean Heat Transport Scenarios

The basal heat flux delivered by the ocean to the ice base varies with latitude under three competing scenarios, parameterised following Lemasquerier et al. (2023):

1. **Uniform transport:**

$$q_{\mathrm{ocean}}(\phi) = q_{\mathrm{mean}} \tag{36}$$

2. **Polar-enhanced** (Lemasquerier et al. 2023):

$$q_{\mathrm{ocean}}(\phi) = q_{\mathrm{mean}} \frac{1 + a \sin^2\phi}{1 + a/3} \tag{37}$$

3. **Equator-enhanced** (Soderlund et al. 2014):

$$q_{\mathrm{ocean}}(\phi) = q_{\mathrm{mean}} \frac{1 + a \cos^2\phi}{1 + 2a/3} \tag{38}$$

The amplitude parameter a is derived from the mantle tidal fraction q^* (Lemasquerier et al. 2023), which measures the fraction of basal heating attributable to tidal dissipation in the silicate mantle ($q^* \in [0, 0.91]$):

$$a = \frac{3\,q^*}{3 - n\,q^*} \tag{39}$$

where $n = 1$ for polar-enhanced and $n = 2$ for equator-enhanced patterns. The normalisation denominators in Eqs. 37–38 ensure that the area-weighted mean heat flux is exactly q_{mean}.

Lateral Heat Diffusion

Explicit lateral diffusion between adjacent columns is applied in geographic coordinates:

$$\left.\frac{\partial T}{\partial t}\right|_{\mathrm{lat}} = \frac{k}{R_p^2 \cos\phi} \frac{\partial}{\partial\phi}\left(\cos\phi \frac{\partial T}{\partial\phi}\right) \tag{40}$$

At the equator ($\phi = 0$), a symmetry condition $\partial T/\partial\phi = 0$ is applied. At the pole ($\phi = \pi/2$), where $\cos\phi \to 0$, L'Hôpital's rule reduces the operator to:

$$\left.\frac{\partial T}{\partial t}\right|_{\text{pole}} = \frac{2\,k}{R_p^2}\,\frac{\partial^2 T}{\partial \phi^2} \tag{41}$$

The thermal diffusion timescale across Europa's radius is $\tau \sim R_p^2/\kappa \sim 200$ Gyr, far exceeding the model's integration window of ∼16 Myr. Consequently, lateral conduction is negligible; the latitude structure is determined entirely by the latitude-varying boundary conditions and tidal forcing.

Monte Carlo Uncertainty Quantification

Forward Propagation

For each Monte Carlo realisation, the model:

1. Draws parameter values from the prior distributions (Table 2).
2. Initialises a 1D ice column with a linear temperature profile.
3. Integrates Eq. 1 to thermal equilibrium using the Stefan condition (Eq. 9) to evolve shell thickness.
4. Records the equilibrium conductive lid thickness D_{cond}, convective sublayer thickness D_{conv}, total shell thickness H, Rayleigh and Nusselt numbers, and all sampled parameter values.

Production ensembles use $N = 5{,}000$–$15{,}000$ realisations, parallelised across CPU cores. Physical filters reject realisations with $H < 0.5\,\text{km}$ or $H > 0.99\,D_{\text{H}_2\text{O}}$ (water layer thickness).

Prior Distributions

Table 2 lists the prior distributions for all sampled parameters, based on the 2026 parameter audit (PARAMETER_PRIOR_AUDIT_2026).

Table 2: Monte Carlo prior distributions (2026 audited).

Parameter	**Distribution**	**Range / Clip**	**Sensitivity**
q_{basal}	Uniform	[5, 25] mW/m^2	Medium
d_{grain}	LogNormal(0.6 mm, $\sigma = 0.35$ dex)	[0.05, 3.0] mm	**High**
ε_0	LogNormal(1.2×10^{-5}, $\sigma = 0.3$ dex)	[2×10^{-6}, 3.4×10^{-5}]	High
μ_{ice}	Normal(3.5 ± 0.5 GPa)	[2, 5] GPa	High
T_{surf}	Normal(104 ± 7 K)	[80, 120] K	Medium
$D_{\text{H}_2\text{O}}$	Normal(127 ± 21 km)	[80, 200] km	Low
Q_v	Normal(59.4 kJ/mol, 5%)	—	**Dominant**
Q_b	Normal(49.0 kJ/mol, 5%)	—	Low
D_{0v}	Normal(9.1×10^{-4}, 3.3%)	—	Low
D_{0b}	Normal(8.4×10^{-4}, 3.3%)	—	Low
H_{rad}	$\lvert\mathcal{N}(4.5, 1.0)\rvert$ pW/kg	> 0	Low
f_{porosity}	Uniform	[0, 0.10]	Negligible
f_{salt}	Fixed at 0	—	—

Equatorial overrides

For equatorial-proxy runs, T_{surf} is replaced by $\mathcal{N}(110\,\text{K}, 5\,\text{K})$ clipped to $[95\,\text{K}, 120\,\text{K}]$, and tidal strain by LogNormal(6×10^{-6}, $\sigma = 0.2$ dex) clipped to $[2\times10^{-6}, 2\times10^{-5}]$. An ocean heat scaling factor $f_{\text{ocean}} \in \{0.55, 0.67, 1.0, 1.2, 1.5\}$ multiplies the tidal component of q_{basal}.

Polar overrides

For polar-proxy runs, T_{surf} is $\mathcal{N}(50\,\text{K}, 5\,\text{K})$ clipped to $[35\,\text{K}, 70\,\text{K}]$, and tidal strain is LogNormal(1.2×10^{-5}, $\sigma = 0.2$ dex) clipped to $[5\times10^{-6}, 5\times10^{-5}]$.

Sobol Sensitivity Analysis

Global sensitivity is quantified via Sobol indices (Sobol' 1993), computed using Saltelli's extension of the Monte Carlo estimator with $N = 512$ base samples. The first-order index S_i measures the fraction of output variance attributable to parameter i alone; the total-order index S_{T_i} includes all interactions involving parameter i.

Bayesian Observational Constraining

The Juno Microwave Radiometer (MWR) provides a constraint on the conductive lid thickness at mid-latitudes: $D_{\text{cond}}^{\text{obs}} = 29 \pm 10\,\text{km}$ (Levin et al. 2026). Rather than re-running the forward model, we apply importance reweighting to the existing Monte Carlo ensemble.

Each realisation i receives a weight:

$$w_i = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{tot}}} \exp\left[-\frac{1}{2}\left(\frac{D_{\text{cond},i} - D_{\text{cond}}^{\text{obs}}}{\sigma_{\text{tot}}}\right)^2\right] \tag{42}$$

where $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{model}}^2}$ combines the observational uncertainty ($\sigma_{\text{obs}} = 10\,\text{km}$) and a model discrepancy term ($\sigma_{\text{model}} = 3\,\text{km}$).

Weighted posterior percentiles are computed via the sorted cumulative weight method:

$$\hat{q}_p = x_{(\pi)} \quad \text{where} \quad \sum_{j\le\pi} w_{(j)} = p \cdot \sum_j w_j \tag{43}$$

The effective sample size (ESS) monitors the quality of the reweighting:

$$\text{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \tag{44}$$

An iterative scheme tightens the priors on q_{basal} and d_{grain} using posterior percentiles, re-runs the ensemble with constrained priors, and iterates until convergence (posterior median within the target range and ESS fraction $> 10\%$).

Bayes factors for model comparison between ocean transport scenarios $\mathcal{M}_j$ are computed as the ratio of marginal likelihoods:

$$\text{BF}_{jk} = \frac{P(D \mid \mathcal{M}_j)}{P(D \mid \mathcal{M}_k)} = \frac{\frac{1}{N_j}\sum_{i=1}^{N_j} w_i^{(j)}}{\frac{1}{N_k}\sum_{i=1}^{N_k} w_i^{(k)}} \tag{45}$$

Summary of Key Physical Constants

Computational Implementation

The model is implemented in Python using NumPy for array operations and SciPy for the tridiagonal solver (`scipy.linalg.solve_banded`). Monte Carlo realisations are parallelised via Python's `multiprocessing` module. Default numerical parameters are listed in Table 4.

Table 3: Planetary and thermal constants.

Parameter	**Symbol**	**Value**	**Unit**
Europa radius	R_p	1,561,000	m
Surface gravity	g	1.315	m/s^2
Orbital frequency	ω	2.047×10^{-5}	rad/s
Eccentricity	e	0.0101	—
Melting temperature	T_m	273	K
Latent heat	L	334,000	J/kg
Reference density	ρ_0	917	kg/m^3
Thermal expansion	α_{th}	1.6×10^{-4}	K^{-1}
Clausius–Clapeyron	$\mathrm{d}T_m/\mathrm{d}P$	-7.4×10^{-8}	K/Pa
Critical Rayleigh no.	$\mathrm{Ra}_{\mathrm{crit}}$	1000	—
Nu prefactor	C	0.3446	—
Nu exponent	ξ	1/3	—
Internal heating exp.	ζ	4/3	—
Lid ratio	θ_{lid}	2.24	—
Andrade exponent	α	0.2	—

Table 4: Default numerical parameters.

Parameter	**Value**	**Description**
n_z	31	Radial nodes per column
N_ϕ	19	Latitude columns (2D mode)
Δt	10^{12} s	Timestep (∼31,700 yr)
$t_{\max}$	10^{15} s	Max integration time (∼31.7 Myr)
Rannacher steps	4	BE half-steps at startup
Picard iterations	3	Max per timestep
Equilibrium tol.	10^{-15} m/s	$\lvert \mathrm{d}H/\mathrm{d}t \rvert$ halt criterion
MC samples	5,000–15,000	Per scenario